

3559. Number of Ways to Assign Edge Weights II

Difficulty: Hard

Link: <https://leetcode.com/problems/number-of-ways-to-assign-edge-weights-ii/description/?envType=daily-question&envId=2026-06-12>

Generated at: 2026-06-12 22:45:13

Problem:

There is an undirected tree with n nodes labeled from 1 to n , rooted at node 1. The tree is represented by a 2D integer array `edges` of length $n - 1$, where `edges[i] = [ui, vi]` indicates that there is an edge between nodes u_i and v_i .

Initially, all edges have a weight of 0. You must assign each edge a weight of either 1 or 2.

The **cost** of a path between any two nodes u and v is the total weight of all edges in the path connecting them.

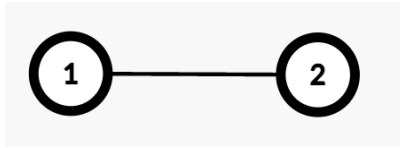
You are given a 2D integer array `queries`. For each `queries[i] = [ui, vi]`, determine the number of ways to assign weights to edges **in the path** such that the cost of the path between u_i and v_i is **odd**.

Return an array `answer`, where `answer[i]` is the number of valid assignments for `queries[i]`.

Since the answer may be large, apply **modulo** $10^9 + 7$ to each `answer[i]`.

Note: For each query, disregard all edges **not** in the path between node u_i and v_i .

Example 1:



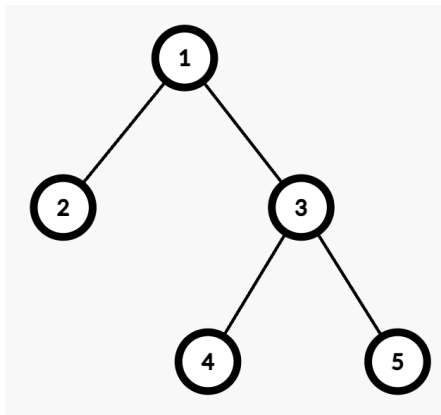
Input: edges = $[[1,2]]$, queries = $[[1,1],[1,2]]$

Output: $[0,1]$

Explanation:

- Query $[1,1]$: The path from Node 1 to itself consists of no edges, so the cost is 0. Thus, the number of valid assignments is 0.
- Query $[1,2]$: The path from Node 1 to Node 2 consists of one edge ($1 \rightarrow 2$). Assigning weight 1 makes the cost odd, while 2 makes it even. Thus, the number of valid assignments is 1.

Example 2:



Input: edges = $[[1,2],[1,3],[3,4],[3,5]]$, queries = $[[1,4],[3,4],[2,5]]$

Output: [2,1,4]

Explanation:

- Query [1,4] : The path from Node 1 to Node 4 consists of two edges (1 → 3 and 3 → 4). Assigning weights (1,2) or (2,1) results in an odd cost. Thus, the number of valid assignments is 2.
- Query [3,4] : The path from Node 3 to Node 4 consists of one edge (3 → 4). Assigning weight 1 makes the cost odd, while 2 makes it even. Thus, the number of valid assignments is 1.
- Query [2,5] : The path from Node 2 to Node 5 consists of three edges (2 → 1, 1 → 3 , and 3 → 5). Assigning (1,2,2), (2,1,2), (2,2,1), or (1,1,1) makes the cost odd. Thus, the number of valid assignments is 4.

Constraints:

- $2 \leq n \leq 10^5$
- `edges.length == n - 1`
- `edges[i] == [ui, vi]`
- $1 \leq \text{queries.length} \leq 10^5$
- `queries[i] == [ui, vi]`
- $1 \leq u_i, v_i \leq n$
- `edges` represents a valid tree.